wisdomikezogwo / quilt1m

Type / to search

Code | Issues 26 | Pull requests 3 | Actions | Projects | Security | Insights

quilt1m

Public

About

[NeurIPS 2023 Oral] Quilt-1M: One Million Image-Text Pairs for Histopathology.

quilt1m.github.io/

#vlm #medical-dataset #multimodal-datasets #histopathology #clip-model

Readme

MIT license

Activity

181 stars

4 watching

9 forks

Report repository

Releases

No releases published

Packages

No packages published

Contributors 1

wisdomikezogwo Ikezogwo oluchi wisdom

Languages

- Python 100.0%

4 Branches 0 Tags

Go to file t | Go to file | Add file | Code

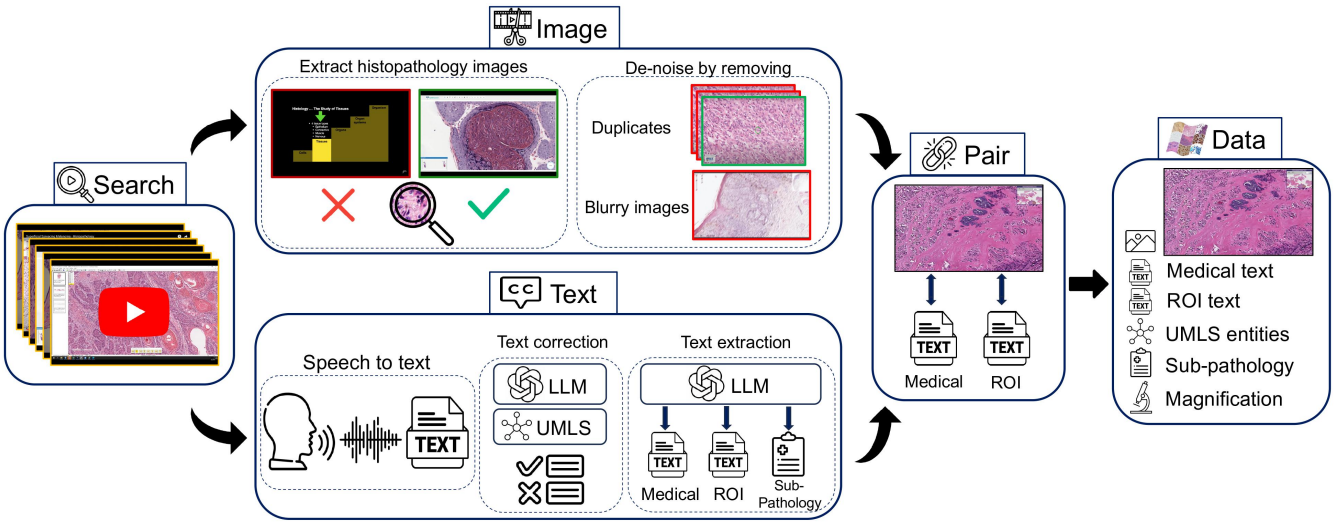
wisdomikezogwo Update README.md 20a5824 · 2 years ago 9 Commits

.idea	recon bug fixes (import and minor)	3 years ago

aux	links and eval added june 25	3 years ago
data	remove sckit-image version	3 years ago
eval	recon bug fixes (import and minor)	3 years ago
.DS_Store	links and eval added june 25	3 years ago
.gitignore	readme update	3 years ago
LICENSE	init	3 years ago
README.md	Update README.md	2 years ago

Quilt

Quilt-1M: One Million Image-Text Pairs for Histopathology [NeurIps 2023] (Oral)



Paper | **Huggingface Demo**

Abstract

> Recent accelerations in multi-modal applications have been made possible with the plethora of image and text data available online. However, the scarcity of similar data in the medical field, specifically in histopathology, has slowed similar progress. To enable similar representation learning for histopathology, we turn to YouTube, an untapped resource of videos, offering 1,087 hours of valuable educational histopathology videos from expert clinicians. From YouTube, we curate Quilt: a large-scale vision-language dataset consisting of 802,148 image and text pairs. Quilt was automatically curated using a mixture of models, including large language models, handcrafted algorithms, human knowledge databases, and automatic speech recognition. In comparison, the most comprehensive

> datasets curated for histopathology amass only around 200K samples. We combine Quilt with datasets, from other sources, including Twitter, research papers, and the internet in general, to create an even larger dataset: Quilt-1M, with 1M paired image-text samples, marking it as the largest vision-language histopathology dataset to date. We demonstrate the value of Quilt-1M by fine-tuning a pre-trained CLIP model. Our model outperforms state-of-the-art models on both zero-shot and linear probing tasks for classifying new pathology images across 13 diverse patch-level datasets of 8 different sub-pathologies and cross-modal retrieval tasks.

News

- ☑ *2023-03-03* Upated repository with links to models and data.
- ☑ *2023-06-13* Initial code/data release.
- ☑ *2023-06-25* Added model evaluate tips and added some new data links.
- ☑ *2023-08-15* Added restricted access to complete dataset.
- ☑ *2023-09-21* QUILT-1M is accepted to NeurIPS 2023 [ORAL] 🔥.
- ☑ *2023-10-26* Corrected sub-pathology column in quilt_1M_lookup csv file, as well as, Added additional columns ('single_wsi': 1-for videos that only cover one WSI or 0-more than one WSI, 'not_histology': manual checks of videos stratified into 0 or 1 with causes of 1 being image projections, drawings or strictly not histo due to false classification videos etc)
- ☑ * *2023-10-27* Updated Arxiv paper.
- ☑ * *2024-01-17* Quilt-LLAVA Paper, Website is released, a Large Language and Vision Assistant for #Pathology trained with spatially localized instruction tuning data generated from educational #YouTube videos, outperforming SOTA in various tasks. Models and data to be released soon.

Data (QUILT-1M) Restricted Access

Two versions of the data can be accessed after agreeing to certain terms, protecting against further distribution of the dataset and committing to its specified research use.

- (Rescaled) On Zenodo you can access the dataset with all images resized to 512x512 px (36 Gb)
- (Full) To access the dataset with full-sized images via Google Drive, please request time-limited access through this form Google (110 Gb)

Requirements

```
conda create --name quilt python=3.9 && conda activate quilt
```

Then install requirements/

Data Reconstruction

To collect Quilt, follow these data steps/

Eval

To evaluate QuiltNet, follow these steps/

README MIT license

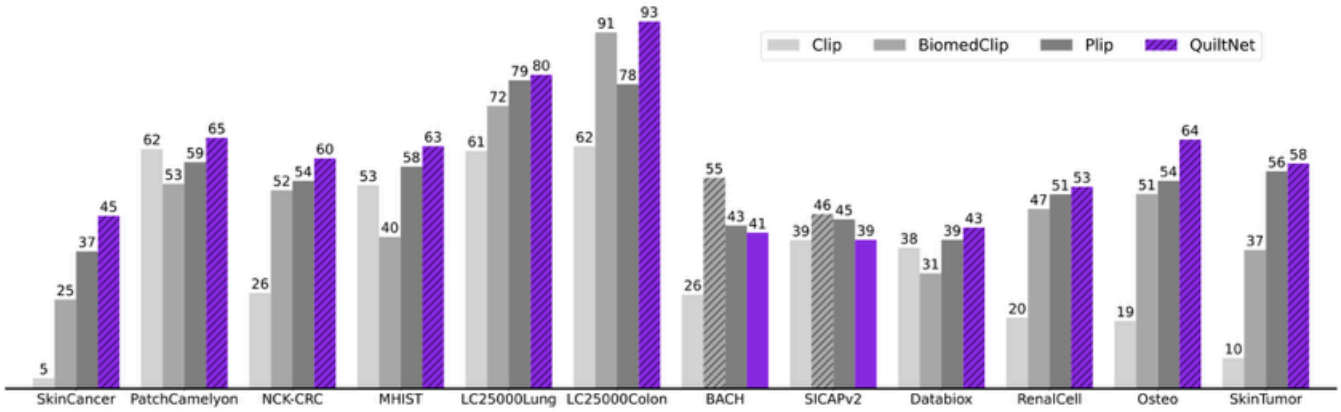
Pretrained Model

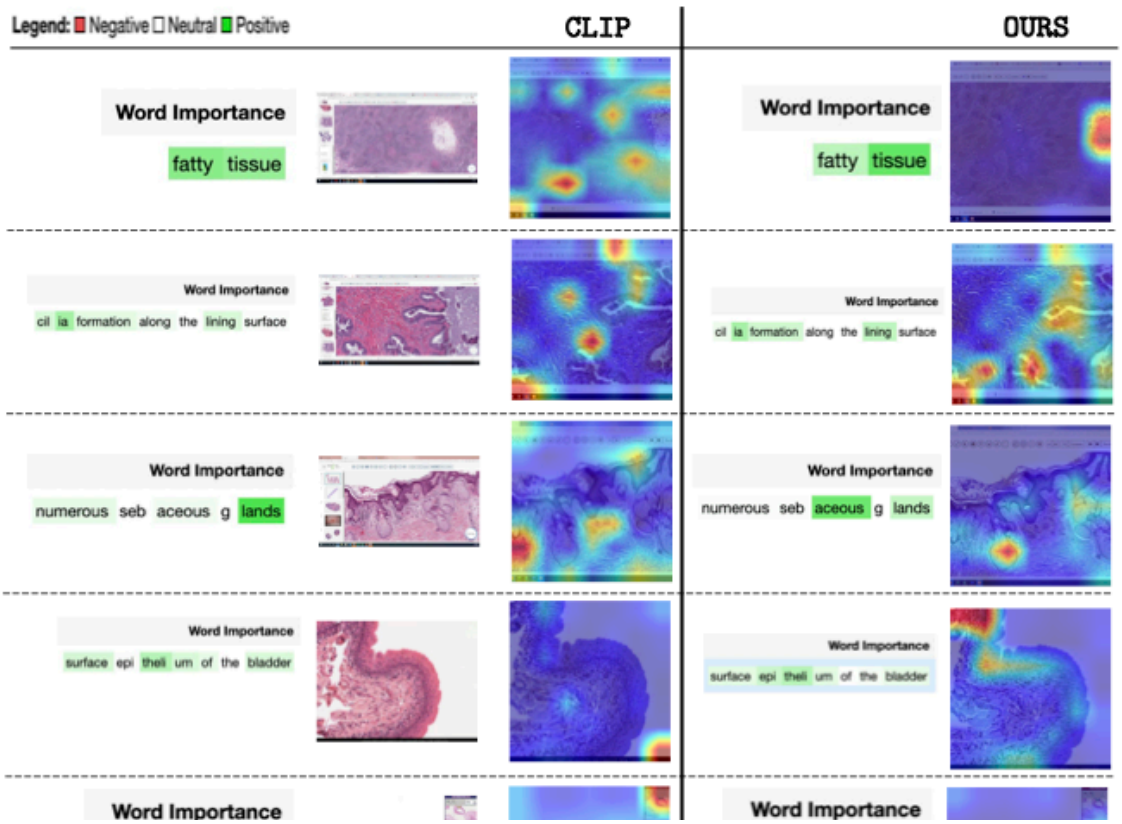
We provide the checkpoints for all QuiltNet finetuned models.

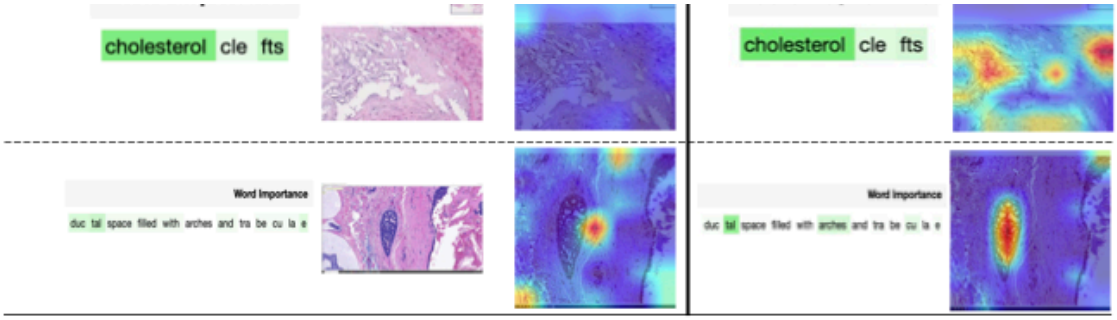
- ViT-B-32|GPT77.
- ViT-B-16|GPT77.
- ViT-B-16|PMB-256.

Testing

Visualization of inputs and output:







Citing Quilt-1M

```
@misc{ikezogwo2023quilt1m,
      title={Quilt-1M: One Million Image-Text Pairs for Histopathology},
      author={Wisdom Oluchi Ikezogwo and Mehmet Saygin Seyfioglu and Fatemeh
Ghezloo and Dylan Stefan Chan Geva and Fatwir Sheikh Mohammed and Pavan Kumar Anand
and Ranjay Krishna and Linda Shapiro},
      year={2023},
      eprint={2306.11207},
      archivePrefix={arXiv},
      primaryClass={cs.CV}
}
```

Acknowledgements

This code borrows heavily from and open-clip and TiMM's library. We also thank the contributors of merlot.

Maintenance

Please open a GitHub issue for any help. If you have any questions regarding the technical details, feel free to contact us.

License

The codes and the pretrained model in this repository are under the MIT license as specified by the LICENSE file.